

CropMind AI: System Architecture Diagram

Layered Microservices, Multi-Modal Feature Synthesis, XGBoost Inference & TreeSHAP Attribution Pipeline

LAYER 1: PRESENTATION & CLIENT INTERFACE TIER

Streamlit Interactive Agronomist Console

- Ranked Top-3 Recommended Crops with % Probabilities
- Interactive What-If Environmental Simulator
- Plotly Interactive SHAP Waterfall & Force Plots
- Agronomic Mitigation & Nitrogen/pH Advisories

Frontend UI

Third-Party REST Clients & Mobile Edge Apps

- Agri-IoT Edge Gateway Terminals (LoRaWAN / ESP32)
- Direct JSON/REST API Client Integration
- Automated Batch Farm Advisory Dispatch
- Standardized Pydantic Data Contracts

External Integrations

LAYER 2: API GATEWAY & APPLICATION MICROSERVICE TIER

FastAPI Production ASGI Microservice (Uvicorn Server)

- POST /api/v1/recommendations/predict (Sub-75ms End-to-End Latency)
- GET /api/v1/health & Readiness / Live Diagnostics
- Pydantic v2 Strict Type Validation & Schema Enforcement
- Automated OpenAPI / Swagger Interactive Documentation

REST Gateway

Spatial Geohash Caching & Weather Synchronization Service

- Level-6 Spatial Geohashing (~1.2 km² Precision Indexing)
- In-Memory Geohash TTL Cache (1-Hour Expiry Window)
- Open-Meteo REST API Connector (Real-Time Rain, T, RH)
- Graceful Offline Fallback to 30-Year Historical Normals

Cache & Sync Engine

LAYER 3: MULTI-MODAL FEATURE SYNTHESIS ENGINE (X in $\mathbb{R}^{12}$)

Stoichiometric Ratio Engine

- N-to-P Ratio: $R_{NP} = N / (P + \epsilon)$
- N-to-K Ratio: $R_{NK} = N / (K + \epsilon)$
- P-to-K Ratio: $R_{PK} = P / (K + \epsilon)$
- Prevents Div-Zero via $\epsilon = 1e-6$

Edaphic Transform

Agro-Climatic Stress Engine

- Temperature-Humidity Index (THI):
 $THI = 0.8 \cdot T + (RH/100) \cdot (T - 14.4) + 46.4$
- Moisture Availability Index (MAI):
 $MAI = (P_{forecast} - \mu_{hist}) / \sigma_{hist}$

Climate Transform

12-D Dense Vector Assembler

- Raw: [N, P, K, T, RH, pH, Rain]
- Synthesized: [R_{NP} , R_{NK} , R_{PK} , THI, MAI]
- Dense Vector Format ($X \in \mathbb{R}^{12}$)
- High-Throughput In-Memory Pipeline

Feature Tensor

LAYER 4: MACHINE LEARNING & EXPLAINABLE AI (XAI) CORE

XGBoost Multi-Class Hist-Tree Classifier (Model Core)

- Histogram-Based Gradient Boosted Decision Forest
- 22 Crop Target Classes (98.86% Test Accuracy | 0.9885 Macro F1)
- Softmax Probability Distribution Vector across all 22 crops
- Ultra-Fast Prediction Time: 5.4 ms Per Inference

Predictive Engine

TreeSHAP Local Factor Attribution & Explanation Engine

- Game-Theoretic TreeSHAP Algorithm (Direct Exact Computation)
- k-Means Centroid Background Reference Set (k=100 Samples)
- Directional Impact Quantifier (+ Δ / - Δ Feature Attribution)
- Causal Agronomic Natural Language Explanation Generator

Explainability Engine

LAYER 5: DATA PERSISTENCE & COLUMNAR STORAGE TIER (ZERO-CSV)

SQLite Relational Database Engine (optic_crop.db)

- 2,200 Indexed Historical Agricultural Samples
- 30-Year Historical Climate Normals (μ_{hist} , σ_{hist} by Region)
- Complete Prediction & SHAP Attribution Audit Logs
- ACID-Compliant Transaction Management (295 KB Footprint)

Relational Storage

Apache Parquet Columnar Data Cache (crop_dataset.parquet)

- High-Performance Vectorized Snappy-Compressed Format (96.5 KB)
- Zero-CSV Architecture: Eliminates CSV Parsing Overhead
- Sub-Millisecond Batch Data Loading for Training & Validation
- Exact Type Preservation (Float32 / Int64 / Categorical)

Columnar Storage

LAYER 6: EXTERNAL SENSORY & METEOROLOGICAL INTEGRATION LAYER

On-Field Edaphic Soil Sensor Telemetry

- Soil Nitrogen (N), Phosphorus (P), Potassium (K) in mg/kg
- Soil Acidity / Alkalinity (pH Scale 0-14)
- Calibrated Sensor Ingestion with Min/Max Clipping Limits
- Real-Time Edge Data Validation

IoT Sensory Data

Open-Meteo REST API Meteorology Provider

- Live Dynamic Ambient Temperature (°C) & Relative Humidity (%)
- 14-Day Dynamic Cumulative Precipitation Forecast (mm)
- Global Geospatial Resolution (No API Key Dependency)
- Average Latency: 38.2 ms Network Request Time

External REST API

CropMind AI: Functional Block Diagram

Signal Acquisition, Feature Synthesis Unit, ML Predictive Core, XAI Attribution Engine & Output Dispatch

